Dr. Eleanor Whitfield, Ph.D.

Department of Computer Science, University of Edinburgh Informatics Forum, 10 Crichton St, Edinburgh EH8 9AB e.whitfield@ed.ac.uk · ORCID: 0000-0002-1234-5678 Scholar: scholar.google.com/citations?user=XXXXXXX · Web: ewhitfield.org

EDUCATION

Ph.D., Computer Science, University of Cambridge, 2019 Dissertation: *Probabilistic Programming for Approximate Bayesian Inference at Scale*. Advisor: Prof. Z. K. Patel.

M.Phil., Advanced Computer Science, University of Cambridge, 2015 — Distinction.

B.A. (Hons), Mathematics, University of Oxford, 2014 — First Class.

ACADEMIC APPOINTMENTS

Lecturer (Assistant Professor), School of Informatics, University of Edinburgh, 2022 – Present.
Postdoctoral Research Fellow, MIT CSAIL, 2019 – 2022. Group of Prof. R. Lin.

RESEARCH INTERESTS

Probabilistic programming languages, approximate inference, machine learning for scientific discovery, and the foundations of variational methods. Particular interest in scaling inference to high-dimensional latent models in computational biology and climate science.

PUBLICATIONS

Peer-reviewed journal articles

1. **Whitfield, E.**, Patel, Z. K., & Lin, R. (2024). Variational gradient descent for hierarchical Bayesian models at scale. *Journal of Machine Learning Research*, 25(78), 1–42.
2. **Whitfield, E.**, & Lin, R. (2023). Amortised inference compilation revisited. *Bayesian Analysis*, 18(2), 401–438.
3. Park, S., **Whitfield, E.**, & Lin, R. (2022). Reproducible probabilistic programs for population genomics. *Genome Biology*, 23, 211.
4. **Whitfield, E.**, & Patel, Z. K. (2020). On the convergence of black-box variational inference under model misspecification. *Statistics & Computing*, 30(4), 945–971.

Conference proceedings

1. **Whitfield, E.**, & Lin, R. (2023). Sparse importance-weighted autoencoders. In *Proceedings of NeurIPS 2023*, 12,341–12,360.
2. Park, S., **Whitfield, E.**, & Lin, R. (2022). Differentiable particle filters for amortised inference. In *Proceedings of ICML 2022*, 18,002–18,019.
3. **Whitfield, E.**, & Patel, Z. K. (2018). Compiling probabilistic programs to accelerator hardware. In *Proceedings of POPL 2018*, 412–429.

Preprints / under review

1. **Whitfield, E.**, Tan, M., & Lin, R. (2025). Score-matching for Bayesian neural networks: theory and scaling. *Under review at JMLR.* arXiv:2502.XXXXX.

GRANTS & FUNDING

- **2024 – 2027** — *Scalable Probabilistic Programming for Climate Science.* UKRI EPSRC. Role: PI. £642,000.
- **2023 – 2025** — *Programming-Language Foundations of Modern Inference.* Royal Society Research Grant. Role: PI. £85,000.
- **2021 – 2022** — *MIT-Lincoln Labs Joint Project on Robust Inference.* Role: Co-I (with R. Lin). $145,000.

AWARDS & HONORS

- **2024** — UKRI Future Leaders Fellowship shortlist.
- **2023** — Outstanding Paper Award, NeurIPS 2023.
- **2019** — Cambridge University Press Best Dissertation Prize, CS Department.
- **2014** — Gibbs Prize for Mathematics, University of Oxford.

INVITED TALKS

- **2024** — “Variational gradients at scale.” Bayesian Computation Workshop, Oberwolfach, Germany.
- **2023** — “Probabilistic programming for population genomics.” Wellcome Trust Sanger Institute, UK.
- **2022** — “Inference compilation revisited.” Statistics Seminar, ETH Zürich.

CONFERENCE PRESENTATIONS

- 2023 — “Sparse importance-weighted autoencoders.” NeurIPS 2023, New Orleans. (Oral)
- 2022 — “Differentiable particle filters.” ICML 2022, Baltimore. (Poster)
- 2021 — “Robust amortised inference under model drift.” UAI 2021, virtual. (Oral)

TEACHING

Term	Course	Institution	Role	Enrollment
2024–25	INFR11102 — Probabilistic Modelling and Reasoning	Edinburgh	Lead instructor	142
2023–24	INFR09019 — Machine Learning Practical	Edinburgh	Co-instructor	96
2021–22	6.S898 — Probabilistic Programming	MIT	Guest lecturer (2 lectures)	38

MENTORING

- **PhD students:** S. O'Connor (2024–, lead supervisor); A. Klein (2023–, co-supervisor with Prof. M. Singh).
- **MSc students supervised:** 5 (2022–2024), all completed; 2 published from thesis work.

SERVICE

- **Reviewer:** NeurIPS (2020–), ICML (2021–), JMLR (2022–), Bayesian Analysis (2023–), Genome Biology (2024–).
- **Area Chair:** UAI 2024.
- **Committee:** Edinburgh Informatics Athena SWAN working group, 2023–present.

PROFESSIONAL MEMBERSHIPS

- International Society for Bayesian Analysis (ISBA), 2019–present.
- Association for Computing Machinery (ACM), 2017–present.

REFERENCES

Prof. Z. K. Patel — PhD advisor University of Cambridge, Department of Computer Science
zk.patel@cam.ac.uk · +44 1223 XXXXXX

Prof. R. Lin — Postdoctoral supervisor MIT CSAIL rlin@mit.edu · +1 617 XXX XXXX

Prof. M. Singh — Faculty colleague (Edinburgh) University of Edinburgh, School of Informatics
m.singh@ed.ac.uk · +44 131 XXX XXXX